UNIVERSITI MALAYSIA SARAWAK (UNIMAS)
Centre For Pre-University Studies

Group Name/Number: Overwatch's Airlines Membership

Group Members & Matric No(s):	**Signatures:**
1. JOSHUA NGO TOM (A2600935)	
2. FERNANADEZ BRYAN ANAK JENNY (A2600717)	
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Subject Code: PRT 1014	**Subject Name: Programming Fundamentals**
Project /Assignment / Lab:	**Final assessment group project**
Name of Lecturer: Encik Khalif Amir Bin Zakry	
Due Date: 21/10/2026	**Date Submitted: 20/10/2026**

All work must be submitted by the due date. If an extension of work is granted, an assignment extension acknowledgement slip must be signed by the lecturer/tutor attached to assignment.

Please note that is your responsibility to retain copies of your assignment.
Plagiarism and Collusion are methods of cheating that falls under Peraturan Akademik Universiti Malaysia Sarawak para 11: Etika Akademik

Plagiarism
Plagiarism isthe presentation of work which has been copied in whole or in part from another person's work, or from any other source such as the internet, published books or periodicals without due acknowledgement given in the text.

Collusion
Collusion is the presentation of work that is the result in whole or in part of unauthorized collaboration with another person or person.

Where there are reasonable grounds for believing that cheating has occurred, the only action that may be taken when plagiarism or collusion is detected is for the staff member not to mark the item of work and to report or refer the matter to the Dean. This may result in work being disallowed and given a fail grade or if the circumstances warrant, the matter may be referred to acommittee of inquiry for investigation. Such investigation may result in the matter being referred to the University DisciplineCommittee, **which** hasthepower toexclude a student.

Artificial Intelligence Tools Usage Declaration Acknowledgement
Haveyou used AI Toolsfor thiswork/assignment/project? ■ Yes ☐ No
If **YES**, how please statefor which parts,why andhowmuch percentage of your total overall work was performed by AI

AI tools were used to assist with reviewing and troubleshooting the Python code. AI was also used to explain Python concepts and logics. AI was used to provide guidance on some harder and newer concepts or logics such as converting str into a date, input validation, optimizing the code logic and improve minor functions.
The base code and original system logic or flow was entirely created on our own while the estimated percentage work performed by AI is 15-30%.

MARK:

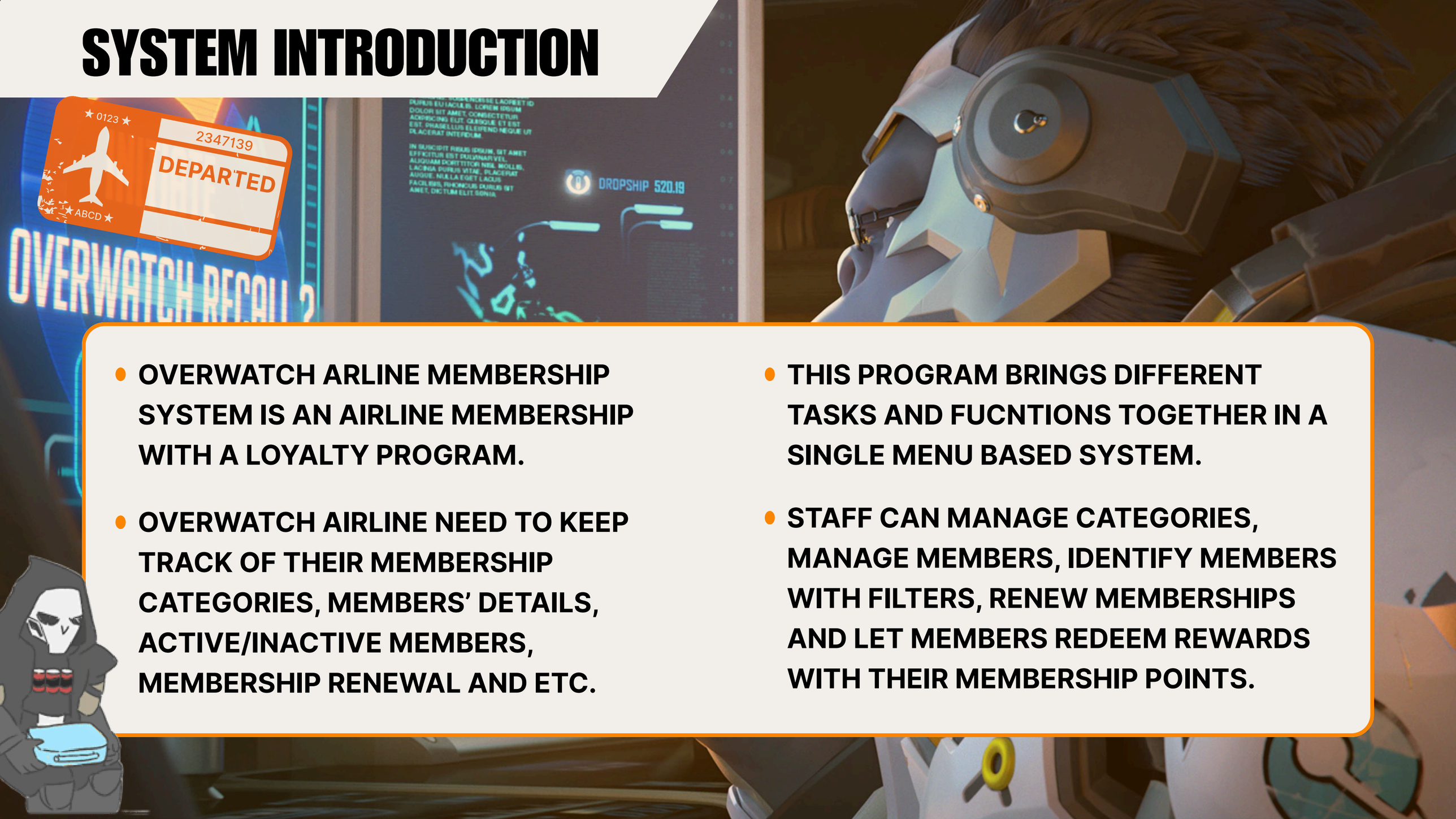
OVERWATCH AIRLINE
MEMBERSHIP SYSTEM



MEMBERS:

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SYSTEM INTRODUCTION



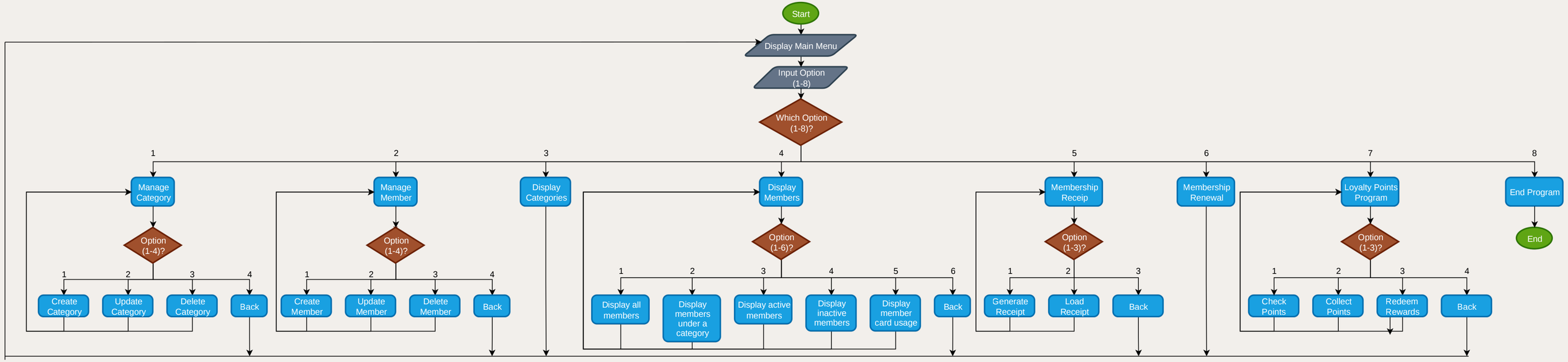
- OVERWATCH ARLINE MEMBERSHIP SYSTEM IS AN AIRLINE MEMBERSHIP WITH A LOYALTY PROGRAM.
- OVERWATCH AIRLINE NEED TO KEEP TRACK OF THEIR MEMBERSHIP CATEGORIES, MEMBERS' DETAILS, ACTIVE/INACTIVE MEMBERS, MEMBERSHIP RENEWAL AND ETC.
- THIS PROGRAM BRINGS DIFFERENT TASKS AND FUCNTIONS TOGETHER IN A SINGLE MENU BASED SYSTEM.
- STAFF CAN MANAGE CATEGORIES, MANAGE MEMBERS, IDENTIFY MEMBERS WITH FILTERS, RENEW MEMBERSHIPS AND LET MEMBERS REDEEM REWARDS WITH THEIR MEMBERSHIP POINTS.

MAIN MENU

1. **Manage Category**
   - Create/Update/Delete Categories
2. **Manage Member**
   - Create/Update/Delete Members
3. **Display Category**
   - Display All Categories
4. **Display Member**
   - Display Member Menu (All Members, Sorted by category, Active/Inactive Members, Member Card Usage)
5. **Generate Member Receipt**
   - Generate/Load Saved Member Receipt
6. **Membership Renewal**
   - Renew Inactive Memberships
7. **Loyalty Points Program**
   - Check member points/Collect earned points/Redeem Rewards
8. **Exit**
   - End the program



FLOW CHART


Start
Display Main Menu
Input Option (1-8)
Which Option (1-8)?
1
2
3
4
5
6
7
8
Manage Category
Option (1-4)?
1
2
3
4
Create Category
Update Category
Delete Category
Back
Manage Member
Option (1-4)?
1
2
3
4
Create Member
Update Member
Delete Member
Back
Display Categories
Display Members
Option (1-6)?
1
2
3
4
5
6
Display all members
Display members under a category
Display active members
Display inactive members
Display member card usage
Back
Membership Receip
Option (1-3)?
1
2
3
Generate Receipt
Load Receipt
Back
Membership Renewal
Loyalty Points Program
Option (1-3)?
1
2
3
4
Check Points
Collect Points
Redeem Rewards
Back
End Program
End

MEMBERSHIP CATEGORIES



SILVER

PERKS:

Extra 10 KG Baggage
Free seat selections
Free Wifi onboard
Priority Boarding

RM 299/year



GOLD

PERKS:

Extra 20 KG Baggage
Airport lounge access
Priority Baggage Handling
Discount for onboard purchases

RM 499/year



DIAMOND

PERKS:

Extra 30 KG Baggage Allowance
Airport lounge access & Guest Passes
Priority Baggage Handling
Discount for onboard purchases

RM 999/year

UNIQUE KEY FEATURE



Loyalty Point Program Feature

- A program that allows members to earn points by flying with Overwatch Airline.
- Points are earned on every flight, calculated by the distance flown.
- This feature allow staff and members to check their current points, collect new points and redeem their rewards using those points.



Python Concepts Used

Dictionaries, Loops, Data Type conversion, Booleans (if, elif, else), Input Validation and Updating Variable.



UNIQUE KEY FEATURE

Check Points	Check & display member’s current total points
Collect Points	Add new earned points of a member
Redeem Rewards	Allow members to redeem rewards if current points are sufficient
Back	Return to main menu



REDEEMABLE REWARDS



MERCHANDISE
500 POINTS



FREE HOTEL STAY
1000 POINTS



FREE FLIGHT TICKET
2500 POINTS



